

Circuit Simulation Lab:13

J-K Flip-flops

Introduction

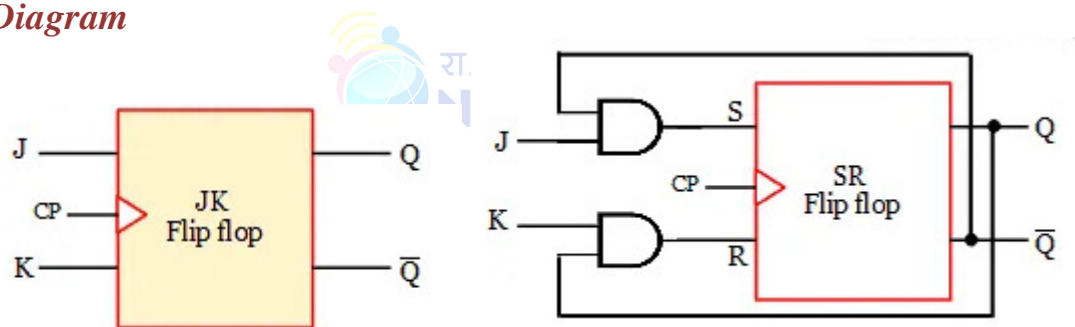
The purpose of this experiment is to introduce the design of simple sequential circuits, in this case J-K Flipflop. JK flip flop is a sequential bi-state single-bit memory element. It has two inputs (J and K), two outputs (Q and Q') and a clock pulse input. It can be triggered either at the positive edge or at the negative edge of the clock pulse.

Software tools Requirement

Modelsim (Siemens)

Xilinx Vivado

Logic Diagram

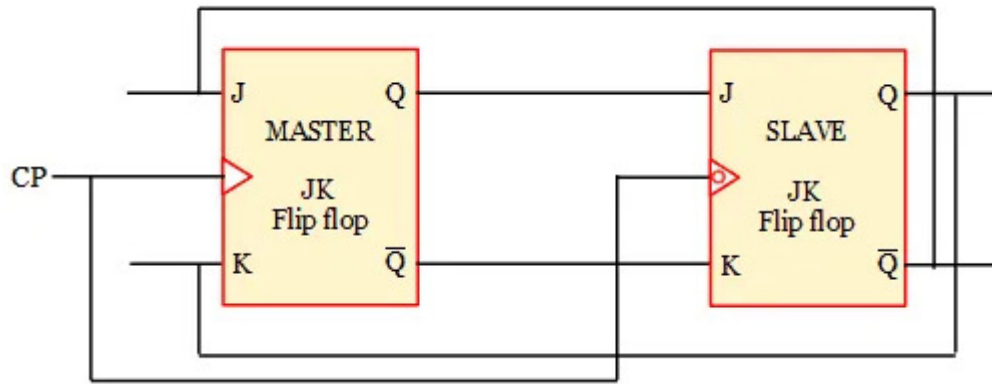


J-K Flip Flop

CP	J	K	Q_{+1}	State
1	0	0	Q_n	NO CHANGE
1	0	1	0	RESET
1	1	0	1	SET
1	1	1	$\bar{Q}_n$	TOGGLES

Truth Table

Master Slave JK Flip-Flop



It consists of two clocked JK flip flops, connected back to back, as shown in the figure below. One flip flop acts as a master and the other flip flop acts as a slave. The input is given to the master flip flop. The output of master JK flip flop is fed as an input to the slave JK flip flop.

Describe the above circuits in Verilog HDL and capture the Waveforms

1. *Behavior modeling*
2. *Structural modeling (Use the JK Flip Flop for designing the MS-JK)*